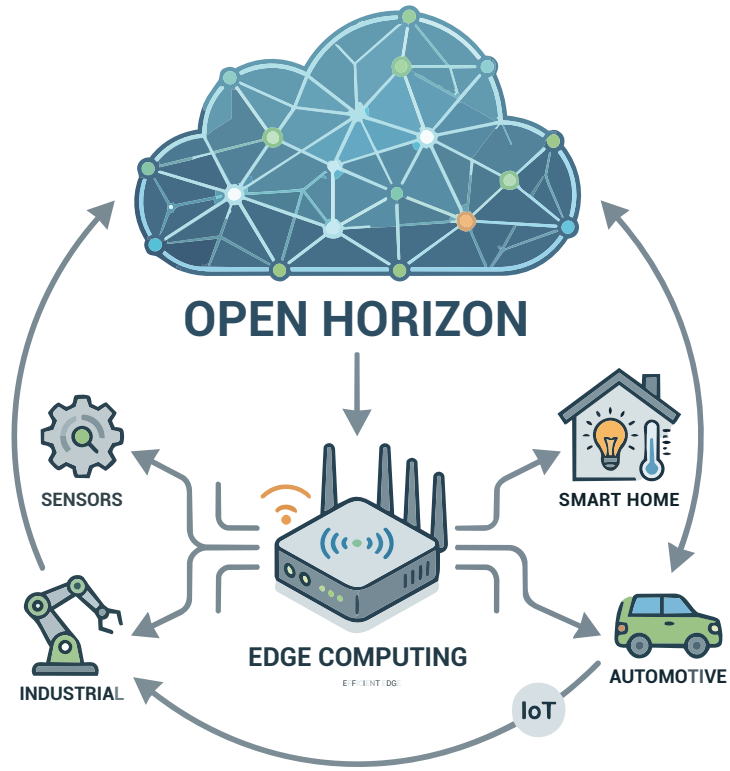
Open Horizon: Enabling Efficient Edge Computing for IoT

Discover why Open Horizon is so useful for IoT edge computing, check its features, learn how to install it; and build, deploy and interact with your first Open Horizon service.



Open Horizon is an open source platform that deploys and manages containerised applications and machine learning models on thousands of edge devices automatically. A project of LF Edge, the Linux Foundation umbrella for edge computing, it began as IBM's Edge Application Manager before IBM contributed the code to open source.

Edge computing moves processing from a distant cloud data centre to the place where data is produced: a factory gateway, a smart camera, a retail kiosk or a Raspberry Pi in a field. For the Internet of Things (IoT), this cuts latency, saves bandwidth and keeps sensitive data local. The hard part is not running one container on one device -- it is keeping the right version of the right software running on 10,000 devices that sit behind firewalls, lose connectivity and have no onsite administrator.

Open Horizon solves that problem. A central management hub stores the services you publish and the rules for where they belong. A small agent on each edge device negotiates with the hub, downloads the containers it should run, starts them and keeps them healthy, even when the network drops.

Why Open Horizon?

Open Horizon is worth choosing when you must run software on many edge devices that you cannot log into one by one. Managing an IoT fleet by hand, with SSH scripts or a cron job per device, breaks down quickly for four reasons.

Scale: Updating 5,000 gateways manually takes days and leaves some devices on old versions. Open Horizon pushes one change to every matching device automatically.

Unreliable networks: Edge devices go offline. The Open Horizon agent works autonomously, keeps services running without the hub, and catches up when the link returns.

Heterogeneous hardware: A fleet mixes x86 servers, ARM boards and Kubernetes clusters. Open Horizon publishes one service per architecture and lets each node pick the version that fits.

No onsite staff: Devices in remote places need self-healing. The agent restarts failed containers and can roll back a bad update.

Compared with sending every reading to the cloud, processing at the edge with Open Horizon lowers latency,